RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Civil Engineering

Date	**09/10/2026**	**Maximum Marks**	**50**
Course Code	**CV232TA**	**Duration**	**90 Min**
Sem	**III**	**CIE – I**	
Environment and Sustainability (Basket Course)			

Sl. No	Questions	M	BT	CO
1.	Classify different types of eco system and explain any two briefly.	10	2	1
2.	Define solid waste management and list the objectives of solid waste management.	10	2	1
3.	i) The design population of a town is 15000. Determine the average daily demand, maximum hourly demand. Assume per capita consumption as 270 LPCD, Maximum daily demand as 1.8 times average daily demand and maximum hourly demand as 2.7 times annual average hourly demand.	05	3	1
	ii) A city generates 250 tonnes/day of municipal solid waste. The compacted density of waste in the landfill is 600 kg/m^3. Calculate the landfill volume required for one year.	05	3	1
4.	i) A municipality with a population of 5,00,000 generates solid waste at 0.55 kg/capita/day. The collection efficiency is 82%. Determine the total waste generated, waste collected and uncollected waste.	05	3	1
	ii) A drainage area consists of roofs, roads and lawns as shown below. If the rainfall intensity is 60 mm/h, determine the overall runoff coefficient and peak runoff.	05	3	1
5.	Write a note on: i) Rain water harvesting ii) Sustainable utilization of energy resources.	10	2	2

Surface	Area (ha)	Runoff coefficient, C
Roofs	6	0.90
Roads	5	0.80
Lawns	9	0.25

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	40	10	–	–	–	30	20	–	–	–

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Department of Civil Engineering

Date: 21/10/2024	**CIE – I (Quiz + Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV232AT**

S No	Questions	M	BT	CO
	Part A			
1.	Distinguish between ecology and ecosystem.	2	1	1
2.	What is primary and secondary succession?	2	1	1
3.	Write any four methods of disposal of solid waste.	2	1	2
4.	What is the importance of safety in workplace?	2	1	2
5.	List any four environmental protection acts in India.	2	1	1
	Part B			
1a.	Briefly explain the structure and components of a ecosystem.	5	2	1
1b.	Materials flow in eco system is bi directional and Energy flow is unidirectional, justify.	5	2	1
2.	With a neat sketch explain the nitrogen cycle.	10	2	1
3.	Define solid waste management and list the objectives of solid waste management.	10	2	2
4.	Define Air Pollution? Classify and explain the different air pollutants and its effects on health and Environment.	10	2	2
5.	Briefly explain the following safety terminologies. i) Accident ii) Ergonomics iii) First aid iv) Hazard v) Safety inspection.	10	2	2

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Department of Civil Engineering

Date: 8th Jan 2024	**CIE – I**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min**
Course Title: Environment and Sustainability		**Course Code: CV232AT**

S No	Questions	M	BT	CO
1a	Define the terms i) Environment ii) Ecology iii) Bio-Diversity.	6	1	1
1b	Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	4	2	1
2	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	10	2	2
3a	Recall the concept of ecological succession and classify the same.	5	1	1
3b	Discuss the approaches of conserving the biodiversity.	5	2	1
4a	Highlight the salient features of various hotspots of biodiversity in India.	5	1	1
4b	Briefly discuss the causes, effects and preventive measures of water pollution.	5	2	2
5	Briefly describe the various Environmental protection acts in India.	10	2	2

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Department of Civil Engineering

Date: 02/04/2025	**CIE – I (Quiz + Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV232AT**

S No	Questions	M	BT	CO
	Part – A			
1	List the components of environment.	2	1	1
2	Define biodiversity.	2	2	1
3	Differentiate between primary and secondary pollutants.	2	3	2
4	The permissible day and night noise levels for a silent zone in India are _____ dB and _____ dB respectively.	2	2	2
5	List any four sources of solid waste.	2	2	1
	Part – B			
1	Classify different types of eco system and explain any two briefly.	10	2	1
2	Define solid waste management and list the objectives of solid waste management.	10	2	2
3	Briefly explain the following safety terminologies. (i) Accident (ii) Ergonomics (iii) First aid (iv) Hazard (v) Safety inspection	10	2	2
4	Define noise pollution. List the effects of noise pollution and write the control measures.	10	2	2
5	With a neat sketch explain carbon cycle.	10	2	1

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Department of Mechanical Engineering

Date: 8th January 2024	**CIE – I**	**Max. Marks: 50**
Semester: 3	**UG**	**Duration: 90 Minutes**
Course Title: Materials Science for Engineers		**Course Code: ME232AT**

S No	Questions	M	BT	CO
1.	Describe all the primary and secondary bonds with examples.	10	2	1
2.	Explain energy bands for metals, Semiconductors, and insulators with a neat sketch.	10	3	1
3a.	Calculate Atomic packing factor for FCC unit cell.	04	4	1
3b.	Define the following with respect to solid materials: i) space lattice, ii) unit cell iii) Atomic Packing Factor.	06	1	1
4.	Explain properties and applications of ceramics and composites with examples.	10	2	1
5.	Explain the following defects: vacancy, edge dislocations, grain boundary.	10	2	1

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Department of Mechanical Engineering

Date: 02/04/2025	**CIE – Test 1**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME242TB**

S No	Questions	M	BT	CO
	Part – A			
1	According to which principle the electrons fill the lowest energy orbital first before moving to higher energy ones.	1	1	1
2	In _____ material the conduction and valence bands overlap so electrons can easily pass into the conduction band.	1	1	1
3	Define unit cell.	1	1	1
4	Write the classification of composites based on matrix.	2	1	1
5	The Burgers Vector is parallel to the dislocation line, the type of line defect is _____.	1	1	1
6	If an atom is missing from its normal site in the matrix, the defect is called _____ defect.	1	1	2
7	Define thermal conductivity with equation.	2	2	2
8	In dielectric materials the movement of electrons in response to an external electric field is _____ type of polarization mechanism.	1	1	2
	Part – B			
1(a)	With the help of neat sketches explain the Primary Interatomic bonds with examples.	10	2	1
2(a)	Classify solid materials explain them briefly with examples.	10	2	1
3(a)	Calculate Atomic packing factor for FCC unit cell.	5	3	2
3(b)	Classify and explain different crystal imperfections.	5	2	2
4(a)	Compare the properties and applications of metals and polymers.	10	3	1
5(a)	Illustrate the following thermoelectric effects: i) Seebeck effect ii) Peltier effect.	5	3	2
5(b)	Discuss different types of Insulating materials highlighting their applications.	5	2	2

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Department of Biotechnology

Date: 22nd October 2024	**Closed Book Offline Quiz & Test – 1**	**Max. Marks: 60**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Quiz)	M	BT	CO
1	Covid 19 virus, will be categorized under which Bio safety level.	1	2	1
2	Mention the composition of Institutional Biosafety Committee.	1	2	2
3	Define pathogen, give example.	1	1	1
4	Mention the purpose of UV lamp in BSCs.	1	2	1
5	List out the working mechanisms of HEPA filters.	2	3	1
6	Expand the terms MERV and ULPA.	2	3	1
7	For the use of GMO at RVCE, what are the approvals required?	2	2	2

S No	Questions (Test)	M	BT	CO
1	Define Biohazard and mention the types of Biohazards with examples for each type.	10	2	1
2	Explain biosafety levels with a neat triangular diagram, with examples for each of the Biosafety level.	10	4	1
3	Illustrate on different types of Biosafety Cabinets, with the help of a neat diagram explain features and advantages of BSC-AII.	10	4	1
4a	Define GMO. Explain with a neat diagram the production of protein rich tomato using carrot gene.	7	2	2
4b	Explain importance and applications of GMOs in various fields.	3	2	2
5a	“Indian farmers, opposed the approval of Genetically Modified mustard seeds in the year 2022”, mention the reasons for opposition and justify the statement in your own words.	7	4	2
5b	Define Cartagena protocol and mention its important functions.	3	2	2

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Department of Biotechnology

Date: January 8, 2024	**Closed Book Offline Test – 1**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Test)	M	BT	CO
1	Elaborate on various bio hazards from nature.	10	2	1
2	List out and discuss biosafety levels with a neat triangular diagram.	10	4	1
3	Explain with a help of a neat diagram, biosafety cabinet B2.	10	4	1
4	Differentiate the bio safety cabinets I, II and III.	10	2	1
5	Give details on GMO and their applications in various fields.	10	2	2

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Department of Biotechnology

Date: 2nd April 2025	**Closed Book Offline Quiz & Test – 1**	**Max. Marks: 60**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Biosafety Standards and Ethics (Basket Course)		**Course Code: BT242TC**

S No	Questions	M	BT	CO
	Quiz			
1	What is the role of the Institutional Biosafety Committee (IBC) in India?	1	2	2
2	The Recombinant DNA Advisory Committee (RCGM) is a part of which Department of Govt of India?	1	1	2
3	The committee of an Institutional Biosafety Committee (IBC) comprise of _____ and _____.	1	1	2
4	Biosafety levels indicate _____.	1	1	1
5	_____ is the primary mode of transmission of many viral biohazards.	1	1	1
6	Which PPE is essential for BSL 4 labs?	1	2	1
7	In class II biosafety cabinet, which direction does the air flow?	1	2	2
8	The Cartagena Protocol was adopted in which year?	1	1	2
9	Mention the 3 types of mechanism which takes place in HEPA filters.	2	3	2
	Test			
1	Comment on various types of Biohazards, and their impact on human health.	10	2	1
2	Explain Biosafety level II class B2 with a neat diagram.	10	2	1
3	Discuss the importance of biosafety guidelines in India and how they contribute to public health and safety.	10	3	2
4	Farmers in the Northern part of India are protesting against the genetically modified Mustard seeds. In this context discuss in detail the pros and cons of the GM Mustard seeds.	10	3	2
5	How do the Biosafety guidelines of the Government of India align with international biosafety standards? Discuss any significant differences or similarities.	10	2	2

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Date: 05th July 2023	**CIE – 1**	**Max. Marks: 25**
Semester: IV	**UG**	**Duration: 60 Min**
Course Title: Environmental Technology		**Course Code: 21BT42A**

S No	Questions	M	BT	CO
1	State the SDG's promulgated by United Nations.	5	2	1
2	Discuss in brief the different components which constitute the environment.	5	3	1
3	State the significance of environmental education and NGO's in conserving the environment.	5	2	1
4	Explain the step by step process of Environmental Impact Assessment with flow chart.	5	2	1
5	Discuss the objectives and applications of ISO 14000 series in Environmental Management System (EMS).	5	2	1